1 Sequences

Definition 1.1. A **sequence** is a function $a : \mathbb{N} \to \mathbb{C}$ (with domain = $\mathbb{N}$, co-domain = $\mathbb{C}$). Instead of denoting the values of the sequence a by $a(1), a(2), \ldots$, we write $a_n = a(n)$ for the n-th term of the sequence. The sequence itself is denoted by $\{a_n\}_{n=1}^{\infty}$, $\{a_n\}$, (a_n), or $(a_1, a_2, a_3, \ldots)$.

Examples

(i) Let $c \in \mathbb{C}$. The sequence $a_n = c$, $n \in \mathbb{N}$, is called the *constant sequence with value* c.

(ii) $a_n = n, n \in \mathbb{N}$. The first terms of (a_n) are $a_1 = 1$, $a_2 = 2$, $a_3 = 3$.

(iii) $a_n = \frac{1}{n}, n \in \mathbb{N}$. The first terms of (a_n) are $a_1 = 1$, $a_2 = 1/2$, $a_3 = 1/3$.

(iv) $a_n = (-1)^n$. All the odd terms of a_n (i.e. a_1, a_3, a_5, etc., generally a_{2k-1}) are equal to -1 and all the even terms of a_n (i.e. a_2, a_4, a_6, etc., generally a_{2k}) are equal to 1.

(v) $a_n = i^n$. The terms repeat in a cycle of four:

$$i, -1, -i, 1, i, -1, -i, 1, \ldots.$$

More precisely, for every $k \in \mathbb{N}$,

$$a_{4k} = 1, \qquad a_{4k+1} = i, \qquad a_{4k+2} = -1, \qquad a_{4k+3} = -i,$$

(vi) $a_1 = 1$ and $a_{n+1} = \sqrt{1 + a_n}$, $n = 1, 2, \ldots$. This sequence is defined *recursively*: if we know a_n, we can calculate a_{n+1} using $a_{n+1} = \sqrt{1 + a_n}$. Since its first term has been given, (a_n) is well defined (we can find a_n in $n - 1$ steps). The first terms of (a_n) are

$$a_1 = 1, \quad a_2 = \sqrt{2}, \quad a_3 = \sqrt{1 + \sqrt{2}}, \quad a_4 = \sqrt{1 + \sqrt{1 + \sqrt{2}}}.$$

(vii)

$$a_n = \begin{cases} \frac{1}{n}, & \text{if } n = 2k, \\ \frac{1}{2}, & \text{if } n = 2k - 1, \end{cases} \qquad k = 1, 2, 3, \ldots$$

To calculate the nth term a_n, it suffices to know whether n is even or odd: for example, $a_6 = 1/6$ and $a_7 = 1/2$.

Remark 1. Notice how general the concept of a sequence is. We can consider any endless list of numbers. It is not necessary that the n-th term be defined by a 'nice formula', as long as we know what each is supposed to be.

Definition 1.2. Let (a_n) and (b_n) be two sequences of complex numbers. We say that $(a_n) = (b_n)$ (the sequences are equal) if $a_n = b_n$ for every $n \in \mathbb{N}$. That is,

$$a_1 = b_1, \quad a_2 = b_2, \quad a_3 = b_3, \ldots.$$

Definition 1.3. The sum, difference, product, and quotient of the sequences (a_n), (b_n) are the sequences $(a_n + b_n)$, $(a_n - b_n)$, $(a_n b_n)$, and (a_n/b_n), respectively (for the last of these, we must make the additional assumption that $b_n \neq 0$ for every $n \in \mathbb{N}$). For $c \in \mathbb{C}$, we define $c(a_n) = (ca_n)$, i.e. we multiply every term with $c \in \mathbb{C}$.

Definition 1.4. The *range* of a sequence (a_n) is the set of values attained by the sequence, namely,

$$A = \{a_n : n \in \mathbb{N}\}.$$

Remark 2. A sequence should not be confused with its range. For example, the sequence $a_n = (-1)^n$ is

$$(-1, 1, -1, 1, \ldots),$$

but its range is simply $\{-1, 1\}$. In particular, different sequences may have the same range.

Definition 1.5. Let (a_n) be a sequence of complex numbers. Any sequence of the form $(a_{m+n-1})_{n=1}^{\infty} = (a_m, a_{m+1}, a_{m+2}, \ldots)$, where $m \in \mathbb{N}$, is called a *tail* of (a_n). For example, the sequences $(5, 6, 7, \ldots)$ and $(30, 31, 32, \ldots)$ are tails of $a_n = n$.

Graphing sequences

Consider the following three sequences:

$$\alpha_n = n, \qquad \beta_n = (-1)^n, \qquad \gamma_n = \frac{1}{n}, \quad n \in \mathbb{N}.$$

Each sequence can be graphed in the coordinate plane by plotting the points (n, a_n).
However, it is often more useful to visualize a sequence on the real number line, marking the values of its terms. In this representation, $\{\alpha_n\}$ consists of the natural numbers and grows without bound, $\{\beta_n\}$ alternates between the two points -1 and 1, and $\{\gamma_n\}$ consists of positive terms that get progressively closer to 0. This way of visualizing sequences helps us understand how their terms behave as n increases.

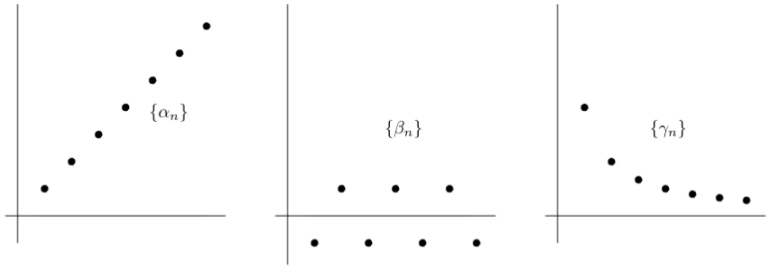


Figure 1: Graphs of the sequences in the coordinate plane

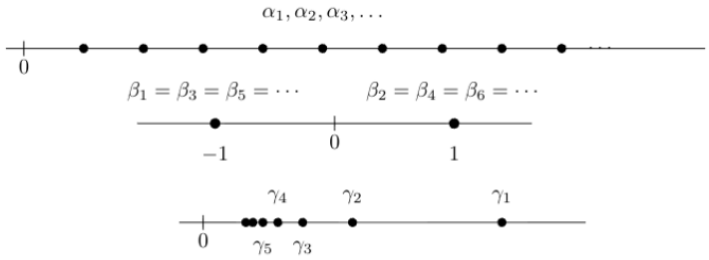


Figure 2: Values of the sequences on the real number line

2 Convergence of sequences

Consider the sequences (a_n) and (b_n) whose n-th terms are

$$a_n = \frac{1}{n} \qquad \text{and} \qquad b_n = (-1)^n.$$

For "large" values of n, the terms $\frac{1}{n}$ of (a_n) are (increasingly) "close" to 0. On the other hand, the terms $(-1)^n$ of (b_n) do not approach any number, as all the odd terms are equal to (-1) while all the even terms are equal to 1 and as a result the sequence oscillates between these two values. We would say that the sequence (a_n) converges (has a limit as n tends to infinity), whereas (b_n) does not converge. In other words, we want to express rigorously the statement:

> "(a_n) converges to a if, for large values of n, a_n is close to a."

What we must clarify is the meaning of the phrases "close" and "large values." For example, if someone considers a distance of 1 to be sufficiently small, then all the terms of (a_n) are close to $1/2$. Also, if someone takes "large values" to mean "quite a few large values," then (b_n) has quite a few terms close to 1, but also quite a few terms close to -1. We agree to say that:

> "(a_n) converges to a if every ball with center a, with arbitrarily small radius, eventually contains all the terms of (a_n)."

Using the notion of a tail of a sequence, we can rewrite this as

> "(a_n) converges to a if for every $\varepsilon > 0$, the ball $B(a, \varepsilon)$ contains some tail of (a_n)."

Observing that $x \in B(a, \varepsilon)$ if and only if $|x - a| < \varepsilon$, we can give the following rigorous definition.

Definition 2.1. Let (a_n) be a sequence of complex numbers and $a \in \mathbb{C}$. We say that (a_n) *converges* to a if the following holds:

> for every $\varepsilon > 0$, there exists a natural number $n_0 = n_0(\varepsilon)$ with the property that, if $n \in \mathbb{N}$ and $n \geq n_0(\varepsilon)$, then $|a_n - a| < \varepsilon$.

If (a_n) converges to a, we write $\lim a_n = a$, or $\lim_{n\to\infty} a_n = a$, or, more simply, $a_n \to a$.
The sequence is called **convergent** if it has a limit. Otherwise, it is called **divergent**.

It is notoriously difficult to appreciate everything packed into the definition of limit. Below are some useful remarks.

Remark 3. Using nested quantifiers we can write that (a_n) converges to a if the following holds:

$$\forall \varepsilon > 0 \ \exists n_0(\varepsilon) \in \mathbb{N} \ : \ |a_n - a| < \varepsilon \ \forall n \geq n_0(\varepsilon).$$

or as

$$\forall \varepsilon > 0 \ \exists n_0(\varepsilon) \in \mathbb{N} \ : \ a_n \in B(a, \varepsilon) \ \forall n \geq n_0(\varepsilon).$$

The geometric interpretation is that a_n is within ε distance of a: it is inside the ball of radius ε centered at a.

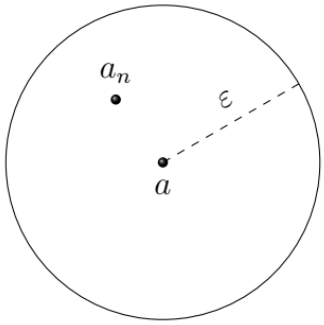


Figure 3: Ball with center a and radius ε.

For a real sequence, $|a - a_n| < \varepsilon$ means $a_n \in (a - \varepsilon, a + \varepsilon) \subset \mathbb{R}$. This interval is the intersection of the ball with the real axis in $\mathbb{C}$.

Remark 4. In the definition above, the index n_0 depends on ε, and for that reason we denote it by $n_0(\varepsilon)$. However small ε may be, we can find $n_0(\varepsilon)$ such that all the terms a_n with $n \geq n_0(\varepsilon)$ are "ε-close" to a. You can think of choosing $n_0(\varepsilon)$ as a challenge against an opponent who keeps choosing smaller and smaller $\varepsilon > 0$.

Remark 5. Given $\varepsilon > 0$ we know that for all $n \geq n_0(\varepsilon)$ $a_n \in B(a, \varepsilon)$. This means that any terms of (a_n) that lie outside of the ball $B(a, \varepsilon)$ must be among $a_1, \ldots, a_{n_0(\varepsilon)-1}$, so only finitely many terms are outside of the ball $B(a, \varepsilon)$. Equivalently, $\lim_{n\to\infty} a_n = a$ means that for every $\varepsilon > 0$, all but finitely many a_n lie in the ball of radius ε centered at a.

Remark 6. The definition of convergence tells us that given $\varepsilon > 0$ there exists a tail $(a_{n_0(\varepsilon)}, a_{n_0(\varepsilon)+1}, \cdots)$ of the sequence (a_n) that lies entirely in $B(a, \varepsilon)$.

Proof template: how to show a sequence a_n converges to a

To prove $\lim_{n\to\infty} a_n = a$ directly from the definition:

Proof. Given $\varepsilon > 0$, let $n_0(\varepsilon) =$ (appropriate expression). Then if $n \geq n_0(\varepsilon)$, we have

$$|a - a_n| = \cdots \leq \cdots < \varepsilon.$$

□

To become familiar with proving such convergence statements using the definition, let us prove that $a_n = \frac{1}{n} \to 0$.

Example 2.1. $\lim_{n\to\infty} 1/n = 0$. Take $\varepsilon > 0$, and consider the open ball $B(0, \varepsilon) = \{z \in \mathbb{C} : |z| < \varepsilon\}$ centered at 0 with radius ε. By the Archimedean property, there exists $n_0(\varepsilon) \in \mathbb{N}$ such that $1/n_0 < \varepsilon$. In particular, we can choose $n_0(\varepsilon) = [1/\varepsilon] + 1$, and then this would be the smallest such natural number. Then, for every $n \geq n_0(\varepsilon)$,

$$\left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{n_0(\varepsilon)} < \varepsilon.$$

That is, for all $n \geq n_0(\varepsilon)$ we have $\frac{1}{n} \in B(0, \varepsilon)$. In other words, the tail $\left(\frac{1}{n_0}, \frac{1}{n_0+1}, \frac{1}{n_0+2}, \ldots\right)$ of the sequence is contained in $B(0, \varepsilon)$. By the definition, $\frac{1}{n} \to 0$.

Remark 7. The choice of n_0 may seem mysterious because proofs are usually constructed backwards on scratch paper. We want

$$\left|\frac{1}{n} - 0\right| < \varepsilon \quad \Longleftrightarrow \quad \frac{1}{n} < \varepsilon \quad \Longleftrightarrow \quad n > \frac{1}{\varepsilon}.$$

This suggests choosing $n_0 = [\frac{1}{\varepsilon}] + 1$. Notice that n_0 has to be a natural number, and with this choice we have $n_0 > \frac{1}{\varepsilon}$, which is exactly what gives $\frac{1}{n_0} < \varepsilon$. Any natural number n with $n > \frac{1}{\varepsilon}$ would also work, so one could invoke the unboundedness of $\mathbb{N}$ to argue that such n exists and use it as n_0.

Example 2.2. Show that $\lim_{n\to\infty}\left(1+\frac{1}{\sqrt{n}}\right)=1$.

On scratch paper:

$$\begin{aligned}\left|1-\left(1+\frac{1}{\sqrt{n}}\right)\right|<\varepsilon &\iff \frac{1}{\sqrt{n}}<\varepsilon\\ &\iff \frac{1}{\varepsilon}<\sqrt{n}\\ &\iff \frac{1}{\varepsilon^2}<n.\end{aligned}$$

Proof. Given $\varepsilon>0$, let $n_0=[1/\varepsilon^2]+1$. Then for $n\geq n_0$ we have, $n>\frac{1}{\varepsilon^2}$ and thus,

$$\left|1-\left(1+\frac{1}{\sqrt{n}}\right)\right|=\frac{1}{\sqrt{n}}<\varepsilon.$$ □

Exercise 2.1. Using the definition of a limit show that

$$\lim_{n\to\infty}\frac{1}{n^2}=0$$

3 Negating the definition of convergence

We are trying to come up with the precise statement of the negation of the definition of a limit. Recall that:

> "(a_n) converges to a if every ball $B(a,\varepsilon)$ with $\varepsilon>0$ contains some tail of (a_n)."

Consequently, (a_n) *does not converge* to a if there exists a ball $B(a,\varepsilon)$ that contains no tail of (a_n). Equivalently,

> "(a_n) does not converge to a if there exists $\varepsilon>0$ such that every tail $(a_m,a_{m+1},\ldots)$ of (a_n) has at least one term that does not belong to $B(a,\varepsilon)$."

Observe that if $(a_m,a_{m+1},\ldots)$ is a tail of (a_n), then $(a_m,a_{m+1},\ldots)$ is not contained in $B(a,\varepsilon)$ if and only if there exists $n\geq m$ such that $a_n\notin B(a,\varepsilon)$, that is, $|a_n-a|\geq\varepsilon$. We thus arrive at the following statement:

> "(a_n) does not converge to a if there exists $\varepsilon>0$ such that for every $m\in\mathbb{N}$ there exists $n\geq m$ with $|a_n-a|\geq\varepsilon$."

It is useful to see how the quantifiers "there exists" and "for all" are flipped when one considers the negation of a statement.
We have $a_n \to a$ if

$$\forall \varepsilon > 0 \ \exists n_0(\varepsilon) \in \mathbb{N} \ \text{s.t.} \ \forall n \geq n_0(\varepsilon) \ |a_n - a| < \varepsilon.$$

Therefore to say $a_n \not\to a$ we negate the above statement as

$$\exists \varepsilon > 0 \ \forall n_0 \in \mathbb{N} \ \exists n \geq n_0 \ \text{s.t.} \ |a_n - a| \geq \varepsilon.$$

Therefore, we obtain the following definition of divergence.

Definition 3.1. The sequence $\{a_n\}$ **diverges** if

$$\forall a \in \mathbb{C} \ \exists \varepsilon > 0 \ \forall n_0 \in \mathbb{N} \ \exists n \geq n_0 \ \text{s.t.} \ |a_n - a| \geq \varepsilon.$$

which in other words means that the sequence does not converge to any number $a \in \mathbb{C}$.

Exercise 3.1. Show that the sequence (a_n) does not converge to a if and only if there exists $\varepsilon > 0$ such that infinitely many terms of (a_n) satisfy $|a_n - a| \geq \varepsilon$.

Example 3.1. Consider the sequence $a_n = (-1)^n$. We will show that (a_n) diverges.

Proof. Let $a \in \mathbb{C}$ be arbitrary. We want to find $\varepsilon > 0$ such that for every $n_0 \in \mathbb{N}$, there exists $n \geq n_0$ satisfying

$$|a_n - a| \geq \varepsilon.$$

Choose $\varepsilon = 1$. By the triangle inequality,

$$2 = |1 - a + a + 1| \leq |1 - a| + |a + 1|.$$

Therefore at least one of the following holds:

$$|1 - a| \geq 1 \qquad \text{or} \qquad |-1 - a| \geq 1.$$

Now let $n_0 \in \mathbb{N}$ be arbitrary. We distinguish two cases:

(i) If $|1 - a| \geq 1$, choose an even integer $n \geq n_0$. Then $a_n = 1$, and hence

$$|a_n - a| = |1 - a| \geq 1 = \varepsilon.$$

(ii) If $|-1 - a| \geq 1$, choose an odd integer $n \geq n_0$. Then $a_n = -1$, and hence

$$|a_n - a| = |-1 - a| \geq 1 = \varepsilon.$$

Thus, for every $n_0 \in \mathbb{N}$, there exists $n \geq n_0$ such that $|a_n - a| \geq \varepsilon$. Since $a \in \mathbb{C}$ was arbitrary, we have shown

$$\forall a \in \mathbb{C} \ \exists \varepsilon > 0 \ \forall n_0 \in \mathbb{N} \ \exists n \geq n_0 \quad \text{such that} \quad |a_n - a| \geq \varepsilon.$$

Therefore, (a_n) diverges. □

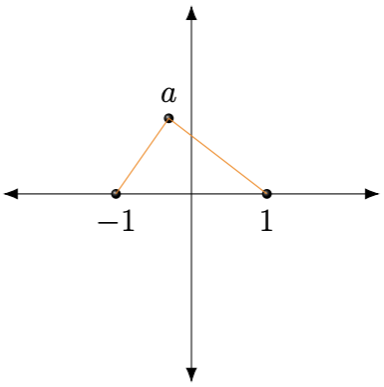


Figure 4: The distance from a to -1 or to 1 has to be at least 1.

Remark 8. Geometrically, there is no point $a \in \mathbb{C}$ that is close to both 1 and -1, i.e. to both the even and the odd terms of the sequence. In the proof, we applied the triangle inequality to the triangle in Figure 4 to conclude that the distance from a to at least one of 1 or -1 must be at least 1. That leads to taking $\varepsilon = 1$. The argument works by taking any ε such that $0 < \varepsilon \leq 1$.

4 Uniqueness of limit

Theorem 4.1. If a sequence converges, its limit is unique.

Proof. Suppose $a_n \to a$ and $a_n \to a'$. Let $\varepsilon > 0$. Applying the definition of convergence with $\varepsilon/2$, we can find natural numbers n_0 and n_1 such that

$$|a_n - a| < \frac{\varepsilon}{2}, \ \forall n \geq n_0$$
$$|a_n - a'| < \frac{\varepsilon}{2}, \ \forall n \geq n_1.$$

Let $N = \max\{n_0, n_1\}$, then we have both $N \geq n_0$ and $N \geq n_1$. Therefore, using the triangle inequality we have

$$|a - a'| = |(a - a_N) + (a_N - a')| \leq |a - a_N| + |a_N - a'| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, the real number $x = |a - a'|$ satisfies $0 \leq x < \varepsilon$ for every $\varepsilon > 0$. By HW6 Problem 1, this gives $x \leq 0$. Hence $|a - a'| = 0$, and therefore $a = a'$. Thus the limit is unique. □